Arjun Mahes

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Education

University of Waterloo Waterloo, ON
B.A.Sc in Mechatronics Engineering, GPA: **3.98/4**

Skills

Languages and Libraries: Python, Java, MATLAB, SQL, C++, PyTorch, Matplotlib, OpenCV
Tools: Solidworks, Git, COMSOL, MS365, KiCad, Altium, STM32CubeMX, Arduino, Bambu Studio, ESP32, Raspberry Pi, Linux

Experience

Engineering Intern May 2026 – Aug 2026
Soneil Spark *Brampton, ON*

- Engineered automated high-voltage AC/DC EV charger testing systems, integrating relay boards, contactors, and simulated EV-to-EVSE communications to validate government safety compliance.
- Developed custom C++ firmware and front-end digital dashboards utilizing Raspberry Pi and ESP32 to autonomously control mentioned testing hardware, reducing test cycle time to 2 minutes per charger and streamlining data collection.
- Constructed and deployed mobile battery trailers in the field, managing complex mechanical builds alongside hardware-firmware-software integration testing.

Hardware Prototyping Engineer Jan 2025 – Apr 2025
Hacker Fab *Waterloo, ON*

- Designed a custom ESP32 control PCB in KiCad with a pressure-sensor input, setpoint knob, LCD readout and stepper driver interface for a PVD sputtering system's argon line.
- Engineered closed-loop control firmware that continuously compares pressure readings to the setpoint and commands a stepper motor to actuate the gas tank valve, regulating argon flow.

Software Engineering Intern Jun 2025 – Aug 2025
DietIQ *Waterloo, ON*

- Architected Retrieval-Augmented Generation (RAG) pipelines and Reinforcement Fine-Tuned models; acquired data by ethically scraping and structuring custom datasets using Selenium.
- Restructured the Python backend architecture for a mobile app to reduce bugs and improve maintainability, and validated API endpoints using Postman.

Projects

Third Thumb Prosthetic | *Altium, Solidworks, KiCad*

- Designed mechanical CAD models in SolidWorks for a robotic third thumb, integrating two servo motors for precise digit articulation and grip control.
- Designed two wireless transmitter PCBs enabling dual-mode proportional control of the thumb: an ESP32-based EMG amplifier with signal filtering, and an nRF52810-driven dual-FSR shoe system translating toe pressure into multi-axis joint articulation.

Ion Trap Simulation | *Python, COMSOL*

- Collaborated in a 4-person team to simulate and optimize Paul and Surface trapped-ion quantum computer geometries in COMSOL Multiphysics, winning 1st Place at the HardHaQ Hackathon.
- Built a Python-COMSOL pipeline using parameter sweeps and SciPy optimizers (Nelder-Mead, SLSQP) to tune geometric and voltage variables against trap depth, trap symmetry, and RF power. **website**
- Designed a novel surface-electrode trap that improved trap depth by 2x and cut estimated RF power by over 30,000x versus the baseline rod design.

Exoskeleton for Stroke Rehabilitation | *Solidworks, Altium, STMCubeMX*

- Designed the electrical system in Altium and the elbow joint mechanical assembly, with an STM32 communicating over SPI, I2C, and UART with a motor driver, magnetic encoder, ESP32, and two EMG sensors
- Implemented Field-Oriented Control for a 3-phase BLDC motor, developing assist-as-needed firmware that dynamically scales torque assistance in inverse proportion to EMG muscle activation, gradually reducing support as patient recovery progresses.
- Developing an EEG-based deep learning model for motor imagery detection to trigger BLDC torque assistance, with ESP32 handling wireless telemetry and command transmission for the brain-computer interface.

Triboelectric Nanogenerator | *Solidworks*

- Designed a rotary triboelectric nanogenerator (TENG) that converts compressed airflow into electricity through contact electrification between PTFE, nylon and aluminum layers, sustaining 30–50 V for over 2 minutes and peaking at 52 V.
- Modelled the full generator assembly in SolidWorks, including the rotor, contact surfaces and air-driven housing, and built it from low-cost household materials.